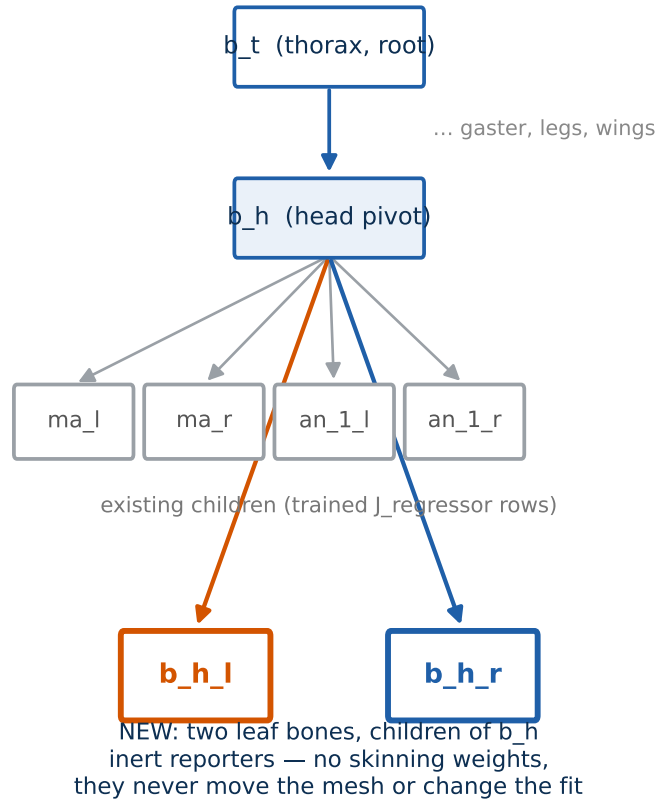


How the head-width joints were added to the model

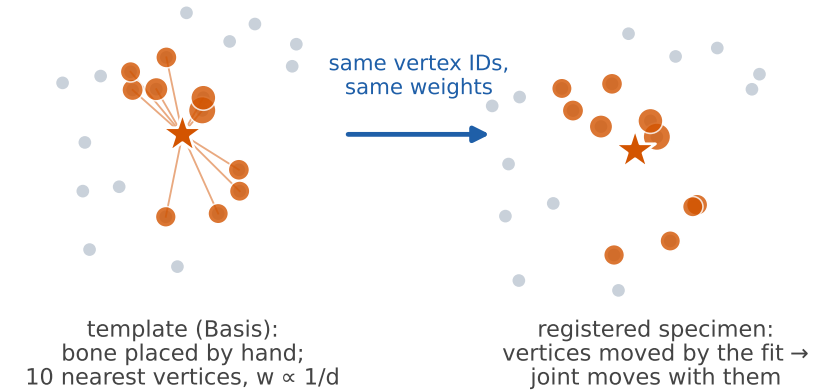
A Position in the SMIL skeleton



B How they were placed (SMIL Model Importer, Blender)

- 1 Import fit**
Direct Import SMIL Model: pkl + ATTA20 npz
PCA / clean_mesh / symmetrise / regress_joints = OFF
- 2 Show the template**
set ALL shape-key values to 0 → Basis mesh only
(exporter reads Basis; keys are additive)
- 3 Mark the widest point**
Object Mode, snap 3D cursor to the head surface
at the widest point of the capsule, full-face
- 4 Add the bone**
Armature Edit Mode → Add ▶ Single Bone at cursor
name b_h_l / b_h_r, parent to b_h
- 5 Export**
Morphometry panel: load body-length CSV →
Export Joint Distances

C What happens on every registered specimen



$$\mathbf{j}_s = \sum_{i \in \mathcal{N}_{10}} w_i \mathbf{v}_{i,s}$$
$$HW_s = \|\mathbf{j}_s^l - \mathbf{j}_s^r\| \times \frac{BL_s^{\text{mm}}}{BL_{\text{model}}}$$

BL = b_t-b_a_5 distance: measured in mm for the specimen, and in model units on the same fit — converts model units to mm

console: [joint distances] J_regressor: 55 trained rows preserved, 2 reporter rows computed (b_h_l, b_h_r)